Main Document

Group 3: Pro Data Scientists **CS 132 WFU**
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Chosen Sustainable Development Goal (SGD):

- ➢ SGD 4 - Quality Education: Education Access, Skills Development, Lifelong Learning

Research Topic: Understanding the Impact of Regional Budgets on the Quality and Accessibility of Higher Education in the Philippines

Research Questions:

The study focuses on investigating how government allocation of regional budgets influences higher education outcomes in the Philippines. Specifically, it explores the following questions:

1. How does regional budget allocation for higher education affect education quality, as reflected by faculty-student ratios?
2. Which regions have higher or lower access to higher education as indicated by enrollment statistics, and how do these disparities relate to regional budget allocations?

Proposed Solutions:

By analyzing data on regional education budgets, enrollment figures, and faculty-student ratios across higher education institutions in the Philippines, this study aims to determine whether higher regional budget allocations are directly correlated with increased access to and quality in higher education. This analysis will provide a clearer understanding of how financial resources influence educational outcomes, while also identifying regions where access and quality may still fall behind others. The insights from this study can guide lawmakers and policymakers in creating data-driven budget allocation strategies that provide more targeted support for regions with lower funding, limited accessibility, or declining education quality. Moreover, this approach promotes more equitable and efficient use of government resources across all regions in the country.

Background:

Access to quality education is essential as it forms the first stage in a person's learning, growth, and future opportunities. In line with this, the Fourth Sustainable Development Goal (SDG 4): Quality Education, emphasizes the need for inclusive and equitable quality education, as well as lifelong learning opportunities. Therefore, it is important to analyze the available data to understand how factors such as regional budget allocation and higher education access are related, and to identify disparities that may hinder the achievement of this goal.

In the Philippines, recent studies have shown that while access to higher education has expanded, disadvantaged students remain less likely to complete higher education and are more often concentrated in lower-status institutions, suggesting that inequalities persist despite reforms (Yee, 2023). At the same time, while education spending per person has grown over the past 25 years, per-student public spending in basic education still lags behind regional peers. For instance, the Philippines spends only about 60–72% of Indonesia's per-student spending despite having a comparable per capita income, which contributes to poor performance in international standardized assessments (Philippine Institute for Development Studies, 2022).

Studying the relationship between government education spending and enrollment in higher education is important for two main reasons. First, it helps determine whether the current use of resources genuinely improves the quality and access to higher education. Second, it highlights regional differences that may persist despite nationwide reforms, which is essential for creating fairer and more effective education policies. Beyond these, examining how education budgets translate into actual outcomes allows us to assess whether investments are aligned with the needs of learners and institutions. It also sheds light on whether policies such as free tuition and scholarships are effective in reducing barriers to higher education. The study aims to contribute to building a stronger evidence base for education reforms, ensuring that government interventions promote long-term development rather than short-term access alone.

Moreover, higher education not only improves individual opportunities but also plays a pivotal role in reducing inequality and fostering economic development. Higher education investments can significantly influence income distribution and economic growth by modifying the circular flow of income (Easdown et al., 2024). Their findings highlight that when individuals, such as working professionals, pursue advanced education, the resulting improvements in income, skills, and productivity strengthen both personal financial stability and national economic performance. This perspective reinforces the idea that educational spending should be viewed not just as a cost, but as a long-term investment in human capital with broad social and economic returns.

By examining these relationships, this study highlights the role of education not only as a basic right but also as a crucial investment in the country's development. The results may guide policymakers in ensuring education budgets are distributed fairly and meet the needs of different regions, in line with the goals of SGD 4.

Hypothesis:

There is a significant relationship between changes in personnel budget allocation and changes in faculty-student ratios across Philippine regions from 2019 to 2024.

There is a significant relationship between enrollment growth patterns and personnel budget allocation changes across Philippine regions from 2019 to 2024.

Description of Data Sets:

1. Higher Education Enrollment Rates and Faculty-Student Ratio per Region (2019 and 2024)
    a. Enrollment data disaggregated by region and institution type (Private Higher Education Institutions, State Universities and Colleges [SUCs], Local Universities and Colleges [LUCs], and Other Government Schools [OGS]).
    b. Faculty–student ratios categorized by sector (Private vs. Public institutions).
    c. *Source:* Commission on Higher Education (CHED), recognized higher education institutions.
2. Higher Education Budget per Region (2019 and 2024)
    a. Budget distribution across categories: Personnel Services (PS), Maintenance and Other Operating Expenses (MOOE), and Capital Outlay (CO).
    b. Data further broken down by central office and regional office allocations.
    c. *Source*: National Expenditure Program (NEP) and Department of Budget and Management (DBM).

Data Collection Process:

The data utilized in this study were obtained from official government documents and statistical releases, specifically those published by the Commission on Higher Education (CHED), the Philippine Statistics Authority (PSA), and the National Expenditure Program (NEP). Relevant tables were extracted from the reports, and the values were manually digitized and encoded into an Excel spreadsheet for analysis. Basic data cleaning was performed, including the standardization of formats and labels, restructuring of tables into a consistent format, and merging of datasets at the regional level.

Data Set Size:

The 2024 Expenditure Program dataset comprises 18 observations, representing the CHED Central Office and the 17 administrative regions, with four budget-related variables. The Higher Education Enrollment and Faculty–Student Ratio dataset includes 18 observations, corresponding to 17 regions and one national total, with seven variables. The 2024 Functional Literacy, Education, and Mass Media Survey (FLEMMS) encompassed 179,560 eligible households and 610,590 individuals aged five years and above, which were subsequently aggregated into 17 regional observations with six variables.

The 2024 Enrollment and Faculty–Student Ratio dataset, comprises 64 observations and 15 variables. This version provides a more comprehensive view of 2024 enrollment patterns, including student counts and the corresponding faculty–student ratios across higher education institutions, both public and private. It serves as a comprehensive source of institutional data for analyzing academic capacity and resource distribution at a detailed level. The same goes for the 2019 dataset, but it includes separate tables for the public and private institutions. For the public institutions, it encompasses 1,540 observations, while private institutions have 114 observations, both with five variables.

Both the 2019 and 2024 Budget dataset from the Department of Budget and Management has 18 regional observations with 5 variables that highlight personnel, maintenance, and capital outlays for the budget.

Overall, the integrated dataset consists of approximately 17 to 18 regional observations, supplemented by entries for the Central Office and national totals, and encompasses between 17 and 20 variables across all data sources for the Enrollment, Faculty-Student Ratio information, Budget.

Data Set: Link to our Data Set - WFU Group 3 - Data Set

Recommendations:

Based on the data analysis, several actionable steps can be recommended to improve both the accessibility and quality of higher education in the Philippines. First, the government and higher education institutions should prioritize addressing the growing faculty-student ratio imbalance through a **targeted faculty hiring initiative**, particularly in regions where ratios have reached critical levels. Strengthening the teaching workforce is essential to maintaining instructional quality as enrollment continues to rise.

Second, budget allocation should be more closely tied to measurable indicators, such as student enrollment growth, faculty-student ratios, and regional demand. By implementing a **needs-based funding** framework, it can help ensure that regions experiencing rapid increases in enrollment receive sufficient support, while preventing under-performance in areas with lower student populations. Alongside this, performance accountability measures should be introduced, requiring universities to regularly report how personnel budgets are used and how these allocations contribute to improving faculty capacity and student outcomes. This approach would ensure that all institutions comply with transparency and efficiency standards in the use of public funds, highlighting accountability within the higher education system.

Third, the government may conduct a **regional equity review** to identify and address funding inconsistencies. Data from this project shows that some high-demand regions, such as Bicol and Ilocos, experienced budget cuts, despite significant enrollment growth, suggesting misalignment between funding and actual needs. A review of these inconsistencies can guide fairer and more efficient distribution of educational resources across these regions.

Furthermore, the project encourages continued **data-driven evaluation of higher education performance**. Integrating future datasets, such as graduation rates, employment outcomes, or licensure exam performance, can provide a more comprehensive review of quality and effectiveness of education provided in the country. This continuous data monitoring approach supports evidence-based policymaking and ensures that future reforms remain responsive to the evolving educational landscape of the higher education system in the Philippines.

Thus, the data analysis and findings gathered call for stronger alignment between funding, faculty resources, and student demand. By making data-informed decisions and promoting equitable allocation, the Philippines can take a critical step toward improving the overall quality, accessibility, and sustainability of its higher education system in the future years.

Conclusion:

This study examined how regional budget allocations affect the quality and accessibility of higher education in the Philippines by comparing data from 2019 and 2024, which is a five-year period. The results revealed a critical issue, wherein Philippine higher education is experiencing a growing quality crisis. The results show that despite substantial increases in both budgets and enrollment, education quality has declined, as shown by the worsening faculty-student ratios, from 22:1 to 31:1. Moreover, the lack of significant correlation between budget changes and faculty hiring or enrollment growth suggests that higher funding alone does not automatically lead to better outcomes.

For the first research question, which explored the relationship between regional budget allocation and education quality, the findings showed no significant correlation between changes in personnel budgets and faculty-student ratios *($r = –0.156$, $p = 0.5643$)*. This suggests that increases in funding have not been effectively utilized to hire more faculty, resulting in inefficient resource use and declining quality of education. In fact, while budgets continued to grow, the average number of students per faculty member also increased, highlighting a contradiction of expanding financial resources but worsening learning conditions.

For the second research question, which studied the relationship between enrollment disparities and budget allocations, the results similarly showed no significant correlation between budget changes and enrollment growth *($r = 0.132$, $p = 0.6251$)*. Although high-access regions experienced large increases in enrollment, such as a 133% rise in Bicol and 112% in Ilocos, many of these regions actually faced budget cuts of up to 17%. Meanwhile, regions with lower or negative enrollment growth, such as the Cordillera Administrative Region, received substantial budget increases. These inconsistencies show inequitable distribution patterns that are not aligned with actual regional demand.

Overall, the findings highlight several systemic challenges within the Philippine higher education system. Budget allocation remains disconnected from both quality needs and student demand. Although regional budgets increased over the five-year period, education quality declined. The lack of a significant correlation between budget changes and faculty-student ratios suggests that additional funding has not effectively improved teaching capacity. This misalignment, combined with insufficient faculty growth relative to enrollment increases, contributes to issues, such as overcrowded classrooms and greater strain on faculty.

Hence, this project highlights the importance of data-driven analysis in addressing structural inefficiencies within higher education. By analyzing regional trends in funding, faculty ratios, and enrollment, it becomes clear that sustainable improvement requires aligning budget allocations with measurable indicators of need and performance. Moving forward, adopting a more transparent, needs-based funding model and ensuring accountability in budget utilization can help build a more equitable and effective higher education system for all regions in the Philippines.

References:

Easdown, N., et al. (2024, June 4). Harmonizing work and higher education: Analysis of the circular flow of income in the Philippine setting. http://dx.doi.org/10.2139/ssrn.4853693

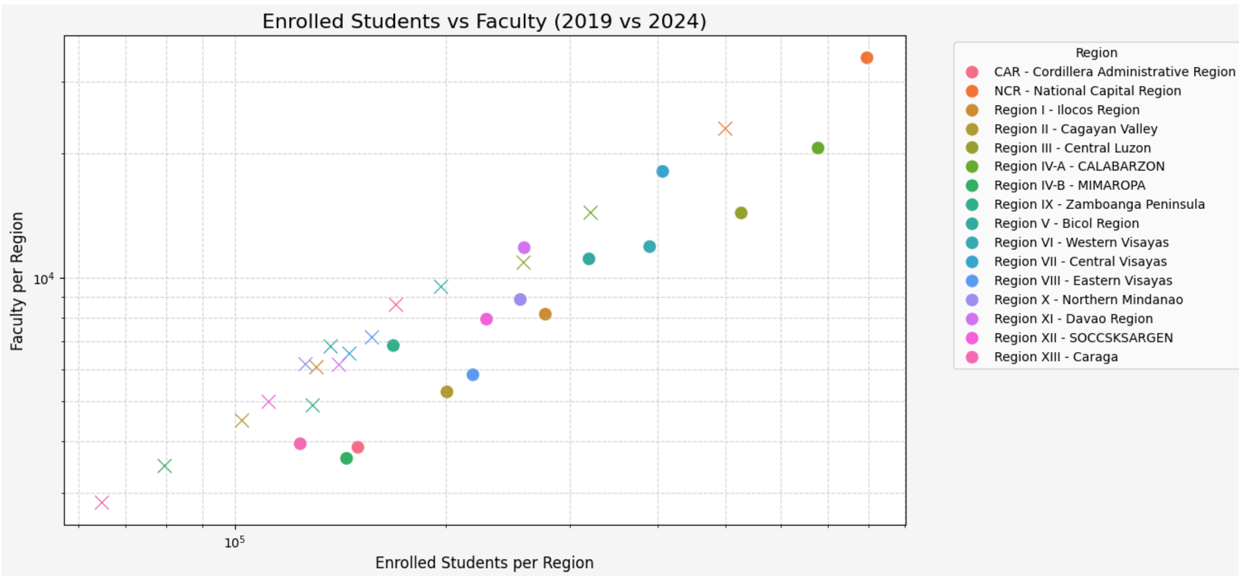
Philippine Institute for Development Studies. (2022, January 22). *PH lags behind regional peers in basic education spending—PIDS study*. https://www.pids.gov.ph/details/ph-lags-behind-regional-peers-in-basic-education-spending-pids-study

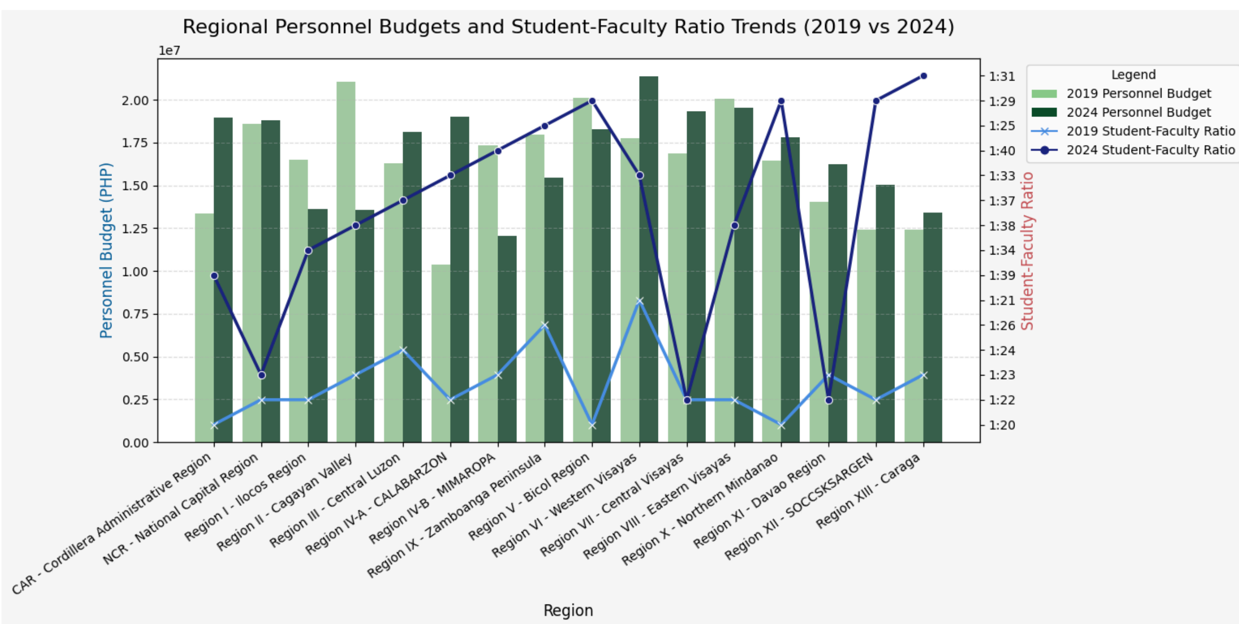
Yee, K. M. R. (2023). At all costs: educational expansion and persistent inequality in the Philippines. *Higher Education*, *87*(6), 1809–1827. https://doi.org/10.1007/s10734-023-01092-y

Graphs

For Research Question 1

How does regional budget allocation for higher education affect education quality, as reflected by faculty-student ratios?





Figures 1 and 2 illustrate regional trends from 2019 to 2024, showing personnel budget allocations alongside faculty-student ratios and the relationship between student enrollment and faculty numbers across regions.

The figures indicate that regional budget allocations for higher education have a significant but uneven impact on education quality, as reflected by faculty-student ratios. Regions with higher personnel budgets, such as the National Capital Region and CALABARZON, show relatively lower student-faculty ratios, suggesting that increased funding can support better faculty availability and smaller class sizes. However, some regions experienced budget increases without notable improvements in faculty-student ratios, indicating that higher funding alone does not automatically translate to improved teaching conditions. These patterns highlight that while budget allocation is an important factor in enhancing education quality, other structural or policy-related factors also influence how effectively resources improve faculty availability and learning conditions.

For Research Question 2

Which regions have higher or lower access to higher education as indicated by enrollment rates, and how do these disparities relate to regional budget allocations?

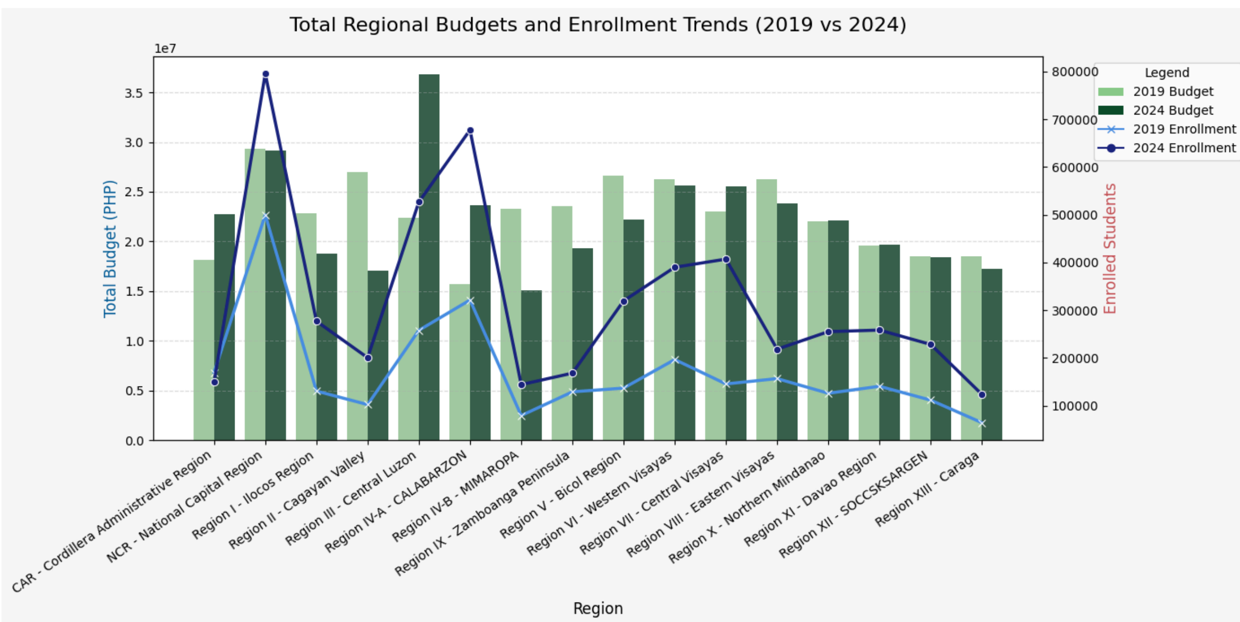
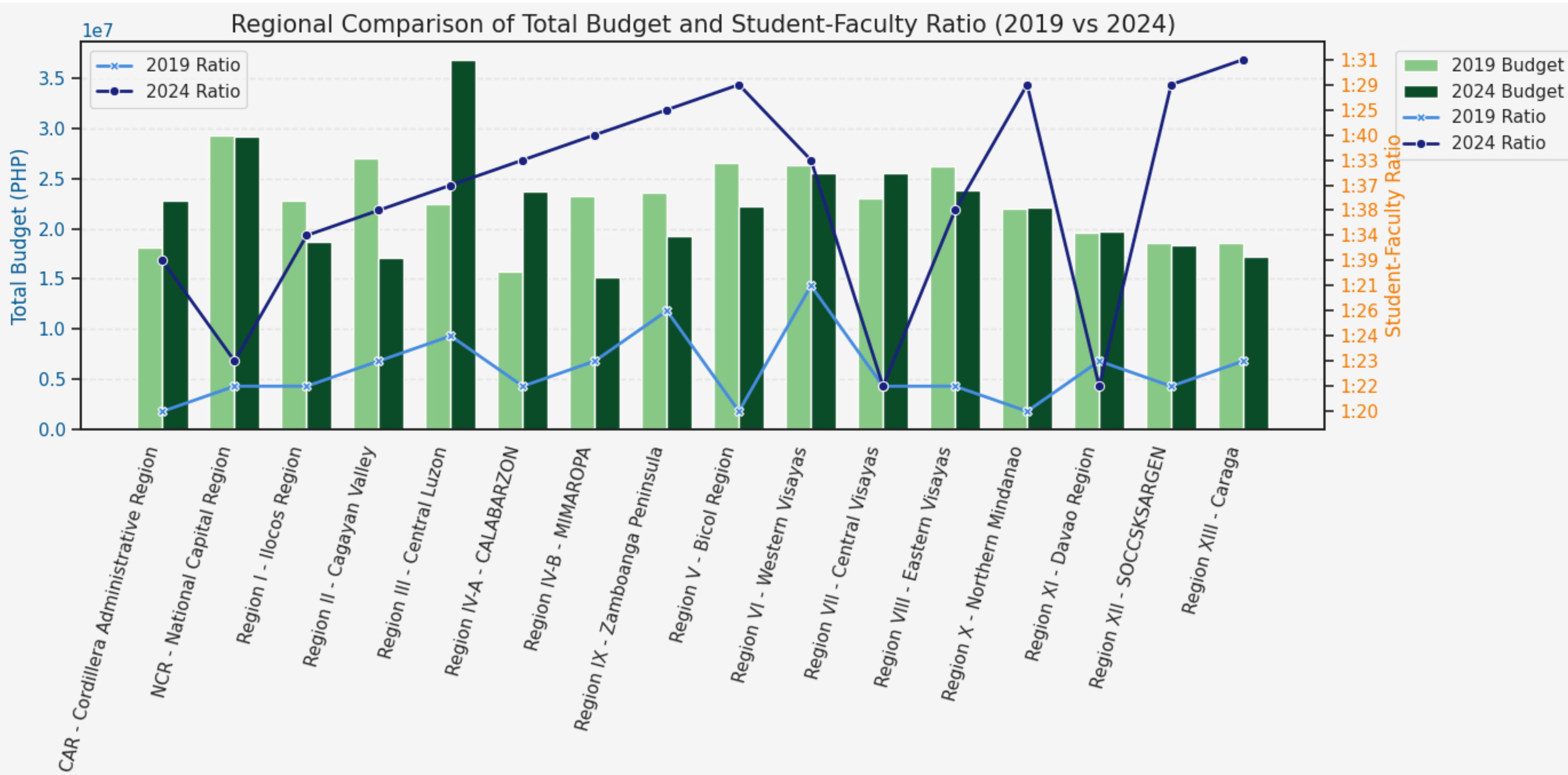


Figure 3 presents the total regional budgets and enrollment trends in higher education from 2019 to 2024. The bar graphs represent total budget allocations in Philippine pesos, while the overlaid lines show the number of enrolled students per region. The figure reveals that regions with higher budget allocations, such as the National Capital Region, Central Luzon, and CALABARZON, also tend to have higher enrollment rates. This pattern suggests that increased financial investment in higher education corresponds with greater student access. In contrast, regions with lower budgets, such as MIMAROPA, Caraga, and the Cordillera Administrative Region, exhibit smaller enrollment figures, indicating possible disparities in access to higher education opportunities. The overall upward trend from 2019 to 2024 suggests that increased government spending has contributed to expanding educational access across most regions.

This supports the hypothesis that higher regional budget allocations are associated with improved access to higher education.

Nutshell Plot



Regional Comparison of Total Budget and Student-Faculty Ratio (2019 vs 2024)
1e7
Total Budget (PHP)
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
Student-Faculty Ratio
1:20
1:22
1:23
1:24
1:26
1:21
1:39
1:34
1:38
1:37
1:33
1:40
1:25
1:29
1:31
2019 Ratio
2024 Ratio
2019 Budget
2024 Budget
CAR - Cordillera Administrative Region
NCR - National Capital Region
Region I - Ilocos Region
Region II - Cagayan Valley
Region III - Central Luzon
Region IV-A - CALABARZON
Region IV-B - MIMAROPA
Region IX - Zamboanga Peninsula
Region V - Bicol Region
Region VI - Western Visayas
Region VII - Central Visayas
Region VIII - Eastern Visayas
Region X - Northern Mindanao
Region XI - Davao Region
Region XII - SOCCSKSARGEN
Region XIII - Caraga

Hypothesis Testing

[GOOGLE COLAB]

Research Question 1: Budget vs. Student-Faculty Ratio

TESTING OUTPUT:

```
H0: No relationship between budget allocation and faculty-student ratios
H1: Significant relationship exists

       *kindly see google colab to see table detailed computation/tables*

Spearman Correlation between Budget Change % and Ratio Change: r = -0.156, p = 0.5643
Result: Fail to reject H0 - No significant relationship

Overall Ratio Change (Paired t-test):
2019 Mean Ratio: 22.17 : 1
2024 Mean Ratio: 31.25 : 1
t = -5.556, p = 0.0001
Significant worsening of faculty-student ratios from 2019 to 2024
```

The analysis found no significant relationship between changes in personnel budgets and changes in student-teacher ratios, indicating that simply increasing funding for personnel has not been an effective strategy for improving class sizes. Despite an average personnel budget increase of 7.24%, the student-teacher ratio worsened significantly across all regions, rising from 22:1 to 31:1, which suggests that budget increases were insufficient to keep pace with massive enrollment growth or were not allocated specifically to hiring new faculty.

Research Question 2: Enrollment Growth vs. Budget Allocation

TESTING OUTPUT:

```
H0: No relationship between enrollment growth and budget allocation
H1: Significant relationship exists

       *kindly see google colab to see table detailed computation/tables*

Spearman Correlation between Enrollment Change % and Total Budget Change %: r = 0.179, p
= 0.5061
Result: Fail to reject H0 - No significant relationship

Budget Equity Analysis:
High access growth regions: 5.42% TOTAL budget change
Low access growth regions: -7.51% TOTAL budget change
TOTAL Budget allocation favors regions with improving access
```

The study revealed no strong statistical link between a region's enrollment growth and its change in total budget, showing that funding has not been systematically proportional to student population increases. However, a positive equity trend was identified, as the regions with the highest enrollment growth received a slight average budget increase of 5.42%, while those with the lowest growth saw their budgets cut by an average of 7.51%, suggesting a partial but weak policy response to access disparities.

SUMMARY